MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

B.Tech. in Electronics Engineering (VLSI Design and Technology)

COURSE PLAN: THEORY COURSE (ODD SEMESTER 2025)

Department:	**ECE**			
Course Name & code:	**Analog Circuits [ECE-2126]**			**Core/PE/OE- Core**
Semester & branch:	**III**	**ECE**		
Name of the faculty:	**Dr. Prashanth Barla**			
No of contact hours/week:	**L**	**T**	**P**	**C**
	3	**1**	**0**	**0**

Course Outcomes (COs/CLOs)

CO/CLO	At the end of this course, the student should be able to:	No. of Contact Hours	Marks	Program Outcomes (PO's)	Learning Outcomes (LOs)*	BL
CO1	Discuss the fundamentals of MOSFET, biasing, characteristics, structure of Op-Amp	7	14	1,2,5	1	2,3,4
CO2	Describe the concept of feedback and oscillators	9	18	1,2,5	2	3,4,5
CO3	Design and analyze MOSFET amplifiers	18	36	1,2,3,5	3	3,4,5
CO4	Design linear circuits, nonlinear circuits, filters using Op-Amp and multivibrators	11	20	1,2,3,5	4	3,4,5
CO5	Analyze the waveform generators and pulse generator circuits	3	10	1,2,4,5	5	5,6
	Total	**48**	**100**			

Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2			2							1
CO2	3	2			2							1
CO3	2	3	3		3							1
CO4	2	3	3		3							1
CO5	2	3		3	2							1
Average Articulation Level	2.4	2.6	1.2	0.6	2.4	-	-	-	-	-	-	1

ICT Tools used in delivery and assessment.

Sl. No	Name of the ICT tool used	Details of how it is used
1	Multimedia Presentations:	PowerPoint / Google Slides with animations of device operation.
2	Whiteboard Tools :	Microsoft OneNote for drawing diagrams live, derivation
3	Simulation Software:	LTspice, Falstad Circuit Simulator (browser-based) for quick demo
4	Learning Management System (LMS):	For Quize, and assignment

Assessment Plan

Component	Type	Weightage	Schedule	Topics		COs
MISAC 1	QUIZ on LMS Platform (Time duration: 30 minutes)	5 Marks	12-08-2025	L1 – L12		1, 3
MISAC 2	Mid-Term Examination	30 Marks		L13 – L23		1,3
MISAC 3	QUIZ on LMS Platform (Time duration: 30 minutes)	5 Marks		L24 – L35		1,2,4
FISAC	Each student will get TEN questions on Self Study topic (Time duration: 30 Days)	10 Marks				1,2,3,4,5

Course Plan

L. No./T. No.	Topics	Course Outcome Addressed
L0	Introduction	CO1
L1	Structure and operation of MOSFET, I–V Characteristics Transconductance	CO1
L2	Channel-Length Modulation, Other secondary effects	CO1
L3	Modelling the MOS Device: Large signal Model	CO1
L4	Modelling the MOS Device:: Small-Signal Model	CO1
L5	PMOS Transistor, CMOS Technology	CO1
L6	MOS Amplifiers: Amplifier topologies, MOSFET Biasing Circuits	CO3
L7	Analysis and Design of Common Source(CS) Amplifier	CO3
L8	CS amplifier with source connected and diode connected load	CO3
L9	CS Stage with Degeneration	CO3
L10	Common-Gate Amplifier	CO3
L11	Source Follower	CO3
L12	Current Mirror, Current Mirror and Active Load	CO3
L13	Current Mirror and Active Load	CO3
L14	Cascode Amplifier	CO3
L15	Multistage CS, CD Amplifiers	CO3
L16	Two Stage CS Amplifiers	CO3
L17	Differential Amplifier: General Considerations	CO3
L18	Qualitative Analysis	CO3
L19	MOS Differential Pair: Large signal anaysis	CO3
L20	MOS Differential Pair: Small signal anaysis	CO3
L21	CMRR	CO3
L22	Differential Pair with Active Load	CO3
L23	Analysis of Two Stage CMOS Operational Amplifier	CO3
L24	Operational Amplifier: Introduction, basic block diagram	CO1
L25	transfer characteristics, measurement of operational amplifier parameters.	CO1
L26	Feedback: Loop Gain, Properties of Negative Feedback, Gain Desensitization	CO2
L27	Bandwidth Extension, Modification of I/O Impedances, Linearity Improvement	CO2
L28	Types of Amplifiers: Simple Amplifier Models, Examples of Amplifier Types	CO2
L29	Feedback Topologies: Voltage-Voltage, Voltage-Current	CO2
L30	Current-Voltage, Current-Current Feedback	CO2
L31	Oscillators: General Considerations, Barkhausen criterion	CO2
L32	Ring Oscillator and LC Oscillators	CO2

L33	Hartley's and Colpitt's Oscillator	CO2
L34	RC Phase Shift Oscillator	CO2
L35	Introduction, Amplifier Fundamentals	CO4
L36	Basic Op-Amp Configurations – Noninverting, Amplifier, Voltage Follower, Inverting Amplifier	CO4
L37	Ideal Op-Amp Circuit Analysis, Summing Amplifier, Difference Amplifier	CO4
L38	Integrator, Differentiator	CO4
L39	Active Filters – First Order Filters, LPF, HPF	CO4
L40	Wide BPF, Phase Shifter	CO4
L41	Nonlinear Circuits – Voltage Comparators, Comparator Applications	CO4
L42	Schmitt Trigger – Inverting and Non-Inverting	CO4
L43	Window Detector and Numaricals	CO4
L44	Precision Rectifiers – HWR, FWR,	CO4
L45	Log and Antilog Amplifier, Analog Multipliers	CO4
L46	Square Wave Generator, Pulse Generator,	CO5
L47	Triangular Wave Generator	CO5
L48	Timer IC 555 – Introduction, Functional Diagram, Astable Operation of IC555, Monostable Operation of IC555	CO5

*Self Directed Learning

DAC types - binary weighted, R-2R. ADC types - flash, successive approximation, dual slope.

Specifications of ADC and DAC.

References:

1. Behzad Razavi, "Fundamental of Microelectronics", Wiley, 2013.

2. A. S. Sedra, K. C. Smith, "Microelectronic circuits", Oxford University Press, 2011.

3. R. L. Boylestad, L. Nashelsky, "Electronic Devices and Circuit Theory", 2009.

4. J. Millman, C. C. Halkias, Chetan. D. Parekh, "Integrated Electronics", McGraw Hill. 2010.

5. Sergio F, Design with Op amps and Analog Integrated Circuits, McGraw Hill, Fourth edition, 2016.

Submitted by: Dr. Prashanth Barla

(Signature of the faculty)

Date:

Approved by: Dr. Pallavi R Mane

(Signature of HOD)